

ECE 100 – Linear Electronics Systems

Lab 2 Report – Active Sallen-Key circuit
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Abstract

This lab analyzed an active low pass filter for separating the digital DSL signal from the voice signal on a telephone line through designing, simulating, and testing. First it started with an analytical estimation step, where a second order Butterworth response was evaluated and then replaced with a second order low pass response that allowed for controlled passband ripple to provide greater design margin without increasing filter order. Then once the damping factor and natural frequency were found, the filter was modified as a Sallen Key active RC circuit and the 4 component values were chosen to achieve the desired response with a gain of -1 dB. The circuit was then simulated using LT Spice, using a LF411 op amp to evaluate non-ideal effects, and the theoretical and simulated magnitude responses were overlaid in MATLAB. Finally, the active circuit was built and experimentally tested by measuring the gain at different frequencies, and compared with the analytical and simulated results.

Experimental Procedures

Part 1- Approximation

For the approximation stage, the filter was designed analytically in MATLAB, a 2nd order Butterworth response was first evaluated by selecting f_0 so that the response just reached the passband edge, then checking whether stopband attenuation at the required frequency met the specification. Because the Butterworth case provided very little margin, a 2nd order low pass response with controlled passband ripple was then used instead. MATLAB was used to then derive and plot the relationship between peak gain and damping factor for values between 0.1 and 0.7, with the values of ζ that gave a 1 dB peak was selected. After that, f_0 was adjusted so the response was -1 dB at 4kHz, and then fixed again so the passband and stopband edges were cleared by approximately the same margin. The theoretical squared magnitude of the frequency response was plotted on a semilog scale over the required frequency range to confirm that the final transfer function met the design goal.

Part 2- Realization/Simulation

In the second part, the transfer function was implemented as a modified Sallen-Key active RC low pass filter. The schematic used for simulation utilized the LF411 op amp, the equivalent resistors, the selected capacitor values, and the voltage divider modification that was used to set the DC gain to -1 dB. The resistor values were originally used from the design equations with R_1 and R_2 both being 100k ohms, and used to find the capacitor values along with the found f_0 and ζ values from the first part. Then LT Spice was used, with an AC simulation to generate the frequency response, with it being checked over the specified range to verify that the simulated circuit still met the pass and stopband band requirements. The resistor values were cut in half and all capacitor values were doubled, to study the non-ideal op amp effects, and the simulation was repeated. The resistor values were then adjusted to find the minimum resistance for which the simulation continued to satisfy the gain requirement, especially at high frequency where the LF411 output resistance becomes important. Then the simulated waveform data was imported into MATLAB so the theoretical and simulated curves of the magnitude of the frequency of the response could be plotted together on the same graph.

Part 3 - Experiment

Finally, the circuit was built and tested using the LF411 op amp chip and the necessary resistor and capacitor values that were calculated in the second part. The function generator was used to connect to the input and output of the circuit so that the gain could be measured directly, and it was measured at dc using 5 Hz, then f_{peak} , 4 kHz, 32kHz, 180 kHz, and 1 MHz. At each of these frequencies, the input and output peak to peak voltages were recorded and converted to gain in dB. At higher frequencies the signal generator amplitude was increased as much as possible, and linear operation was confirmed by changing the input amplitude by a factor of 2 and the gain was measured again. If the gain remained the same, the circuit was assumed to be operating linearly; the measured data was then saved and imported into MATLAB on top of the theoretical and simulated responses for final comparison.

Results

Part 1- Approximation

- (a) The magnitude squared of the frequency response of a Butterworth filter is $|H(f)|^2 = \frac{1}{1+(\frac{f}{f_0})^4}$. To see if the transfer function meets the spec, we choose f_0 so the transfer function just hits the passband edge. This is the lowest possible value of f_0 . Then we check to see if $|H(f)|^2$ clears the stopband edge. Find this value of f_0 analytically. Calculate the resulting $|H(f)|^2$ at 32 kHz and 180 kHz. Does the filter meet the spec?

a)

We design a low pass filter we will use
Second Order RLC low-pass

from Lab 11:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

For a Second Order Butterworth Circuit:

$$\frac{1}{1+(\frac{f}{f_0})^4} = |H(f)|^2$$

from the graph we know:

$$-1 \text{ dB} = |H(4 \text{ kHz})|^2$$

$$10 \log_{10}(|H(4 \text{ kHz})|^2) = -1 \text{ dB}$$

$$\log_{10}\left(\frac{1}{1 + \left(\frac{4 \text{ kHz}}{f_c}\right)^4}\right) = \frac{-1}{10} \text{ dB}$$

$$10 \log_{10}\left(\frac{1}{1 + \left(\frac{4 \text{ kHz}}{f_c}\right)^4}\right) = 10 \left(\frac{-1}{10}\right)$$

$$\frac{1}{1 + \left(\frac{4 \text{ kHz}}{f_c}\right)^4} = 0.7943$$

$$1 + \left(\frac{4 \text{ kHz}}{f_c}\right)^4 = \frac{1}{0.7943}$$

$$\frac{4 \text{ kHz}}{f_c} = (0.258925)^{1/4}$$

$$\frac{4 \text{ kHz}}{f_c} = 0.713335227 \rightarrow f_c = \left(\frac{0.713335}{4 \text{ kHz}}\right)^{-1} = 5.60746 \text{ kHz}$$

$$|H(f)|^2 \rightarrow 32 \text{ kHz}$$

$$|H(32 \text{ kHz})|^2 = \frac{1}{1 + \left(\frac{32 \text{ kHz}}{5.6075 \text{ kHz}}\right)^4} = 10 \log_{10}(9.4 \times 10^{-4}) = -30.25 \text{ dB} \rightarrow \text{clear spec: } < -30 \text{ dB}$$

* Go through Lectures 6-7 otherwise won't have knowledge to do lab!

$$|H(180 \text{ kHz})|^2 = \frac{1}{1 + \left(\frac{180}{5.607}\right)^4} = 10 \log_{10}(9.4 \times 10^{-7}) = -60.25 \text{ dB} \rightarrow \text{clear spec: } < -55 \text{ dB}$$

- (b) You will find that the maximally flat transfer function (the characteristic of the Butterworth filter) does meet the spec, but it is a very tight fit. This leaves little room for component errors. We could increase the order from $n = 2$ to $n = 3$, but this will increase the cost by, roughly, 50%. So instead, we will design and use another filter transfer function. We will allow the $|H(f)|^2$ in the passband to “ripple” from -1dB at dc up to 0dB, then down to -1dB at 4 KHz. Thus, $|H(f)|^2$ will have a peak 1dB greater than the dc value. At dc,

$$10 \log |H(f = 0)|^2 = -1dB, \text{ which means } |H(f = 0)|^2 = 0.7943.$$

Following the standard transfer function for a second-order low-pass filter with DC gain of -1dB, $H(s) = \frac{(0.7943)^{\frac{1}{2}}}{1 + \frac{2\zeta s}{\omega_0} + \frac{s^2}{\omega_0^2}}$, find an expression for the frequency at which $|H(j\omega)|$ peaks. Also, find an expression for $\frac{|H(f_{peak})|^2}{|H(0)|^2}$ as a function of ζ . Use MATLAB to plot $\frac{|H(f_{peak})|^2}{|H(0)|^2}$ in dB vs ζ in the range $0.1 < \zeta < 0.7$ and find the value of ζ that will provide a 1dB peak.

- b) pass band ripple -1dB up to 0dB
 -1dB at 4kHz

$|H(f)|^2$ has peak 1dB greater than dc value ($f=0$ kHz)

$$10 \log_{10} |H(f=0)|^2 = -1 \text{ dB}$$

$$|H(f=0)|^2 = 0.7943$$

Approximate line $\Rightarrow$ -40 dB/decade for $\omega \gg \omega_0$
 -12 dB/octave

$$H(s) = \frac{(0.7943)^{1/2}}{1 + \frac{2\zeta s}{\omega_0} + \frac{s^2}{\omega_0^2}} \quad \Bigg|_{s=j\omega} \quad H(j\omega) = \frac{(0.7943)^{1/2}}{1 + \frac{2\zeta(j\omega)}{\omega_0} + \frac{(j\omega)^2}{\omega_0^2}}$$

$$|H(j\omega)| = \left| (0.7943)^{1/2} \cdot \frac{1}{1 - \frac{\omega^2}{\omega_0^2} + j \frac{2\zeta\omega}{\omega_0}} \right|$$

$$= \frac{(0.7943)^{1/2}}{\sqrt{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_0}\right)^2}}$$

$$\left| \frac{1}{a+jb} \right| = \frac{1}{\sqrt{a^2+b^2}}$$

$$|H(j\omega)|^2 = \frac{(0.7943)}{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + \left(\frac{2\zeta\omega}{\omega_0}\right)^2}$$

↳ Cont.

$$\frac{d}{d\omega} |H(j\omega)| = 0 \quad \text{to find maxima}$$

↳ we derived for a Second Order RLC Low pass filter in Lab 1:

$$\omega_{peak} = \omega_0 \sqrt{1 - 2\gamma^2}$$

↓

$$\text{we know } f = \frac{\omega}{2\pi}$$

↓

$$\text{Hence form: } f_{peak} = \frac{\omega_{peak}}{2\pi} = \frac{\omega_0 \sqrt{1 - 2\gamma^2}}{2\pi} = f_0 \sqrt{1 - 2\gamma^2}$$

Peak equation

$$\begin{aligned} |H(j\omega_{peak})| &= \frac{(0.7943)^{1/2}}{\sqrt{\left(1 - \frac{\omega^2}{\omega_0^2}\right)^2 + \left(\frac{2\gamma\omega}{\omega_0}\right)^2}} \quad \left| \omega_{peak} = \omega_0 \sqrt{1 - 2\gamma^2} \right. \\ &= \frac{(0.7943)^{1/2}}{\left(\left(1 - \frac{\omega_0^2 \sqrt{1 - 2\gamma^2}}{\omega_0^2}\right)^2 + \left(\frac{2\gamma \omega_0 \sqrt{1 - 2\gamma^2}}{\omega_0}\right)^2 \right)^{1/2}} \\ &= \frac{(0.7943)^{1/2}}{\left(\left(1 - \sqrt{1 - 2\gamma^2}\right)^2 + \left(2\gamma \sqrt{1 - 2\gamma^2}\right)^2 \right)^{1/2}} \\ &= \frac{(0.7943)^{1/2}}{\left(1 - 2\sqrt{1 - 2\gamma^2} + 1 - 2\gamma^2 + 4\gamma^2(1 - 2\gamma^2) \right)^{1/2}} \\ &= \frac{(0.7943)^{1/2}}{\left(4\gamma^4 + -8\gamma^4 + 4\gamma^2 \right)^{1/2}} \\ &= \frac{(0.7943)^{1/2}}{\sqrt{4(\gamma^2 - \gamma^4)}} \\ &= \frac{(0.7943)^{1/2}}{2\gamma \sqrt{1 - \gamma^2}} \end{aligned}$$

$$|H(f_{\text{peak}})| = |H(j \frac{\omega_{\text{peak}}}{2\pi})| = \frac{(0.7943)^{1/2}}{\sqrt{\left(1 - \frac{\omega_p^2}{(2\pi)^2 \omega_0^2}\right)^2 + \left(\frac{2\gamma \omega_p}{2\pi \omega_0}\right)^2}} \quad \left| \omega_p = \omega_0 \sqrt{1-2\gamma^2} \right.$$

$$= \left[\left(1 - \frac{\omega_0 \sqrt{1-2\gamma^2}}{(2\pi)^2 \omega_0}\right)^2 + \left(\frac{2\gamma \omega_0 \sqrt{1-2\gamma^2}}{2\pi \omega_0}\right)^2 \right]^{1/2}$$

$$|H(f_{\text{peak}})|^2 = \frac{0.7943}{\left(1 - \frac{(1-2\gamma^2)}{(2\pi)^2}\right)^2 + \left(\frac{2\gamma}{\pi} \sqrt{1-2\gamma^2}\right)^2}$$

$$= \frac{0.7943}{\left(1 - \frac{(1-2\gamma^2)}{(2\pi)^2}\right) + \frac{\gamma^2}{\pi^2} (1-2\gamma^2)}$$

$$= \frac{(0.7943)}{(2\gamma)^2 (1-\gamma^2)} = \frac{0.7943}{4\gamma^2 (1-\gamma^2)}$$

$\frac{|H(f_{\text{peak}})|^2}{|H(0)|^2}$ as a function of γ

$$\frac{|H(f_{\text{peak}})|^2}{|H(f=0)|^2} = \frac{\frac{0.7943}{4\gamma^2 (1-\gamma^2)}}{0.7943}$$

$$= \frac{1}{4\gamma^2 (1-\gamma^2)}$$

matlab plot shows $\gamma \approx 0.523$ provides a 1dB peak

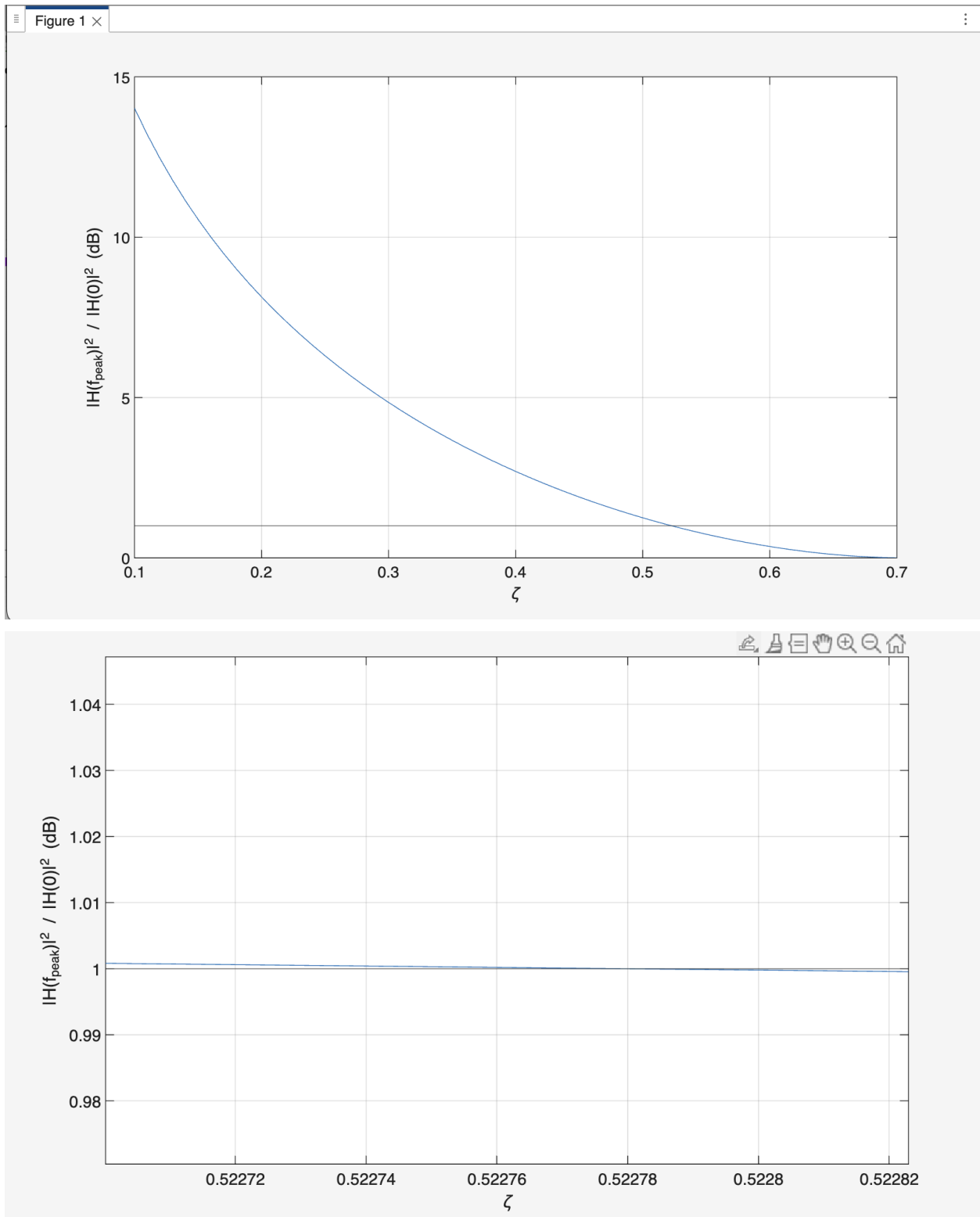


Figure 1: Matlab Plot depicting value of ζ that provides peak of 1dB

(c) Now that we have the necessary ζ , we can correct the dc gain using a voltage divider to get $|H(f)|^2 = \frac{0.7943}{(1-(\frac{f}{f_0})^2)^2 + (2\zeta\frac{f}{f_0})^2}$. All that remains is to choose the natural frequency f_0 so $|H(f)|^2 = -1\text{dB}$ at 4 KHz. It is not easy to invert this equation for f_0 either, although MATLAB can do that. One possible way to find f_0 is as follows. Take a guess at f_0 , say $f_0 = 5000\text{Hz}$, and plot $|H(f)|^2$ in dB vs f over the range $1,000 < f < 10,000$. Find the frequency at which $|H(f)|^2$ drops to -1dB, say this value is x . We therefore need to scale f_0 by the factor $\frac{4000}{x}$, i.e. $f_0 = 5000 \times \frac{4000}{x}$. Find the required f_0 and plot $|H(f)|^2$ in dB vs f for $200 < f < 1\text{MHz}$ on a semilogx plot. The resulting function will drop faster in the transition band and will meet the spec with more clearance than the Butterworth approximation. There is no need to save a copy of this plot.

c) Choose f_0 such that:

$$|H(f)|^2 = -1\text{dB at } f = 4\text{kHz} \quad |H(f)|^2 = \frac{0.7943}{(1-(\frac{f}{f_0})^2)^2 + (2\zeta\frac{f}{f_0})^2}$$

$$10 \log_{10} |H(4\text{kHz})|^2 = -1\text{dB}$$

$$\frac{0.7943}{(1-(\frac{4000}{f_0})^2)^2 + (2(0.523)\frac{4000}{f_0})^2} = 10^{-1/10}$$

pretend $f_0 = 5000\text{Hz}$

plot $10 \log_{10} |H(f)|^2$ in dB vs. f for $1,000 < f < 10,000$

find $10 \log_{10} |H(f)|^2 = -1\text{dB}$ $f = x$

$$f_0 = 5000 \cdot \frac{4000}{x}$$

$$f_0 = 5000 \cdot \frac{4000}{4736.5} = 4222.53\text{Hz}$$

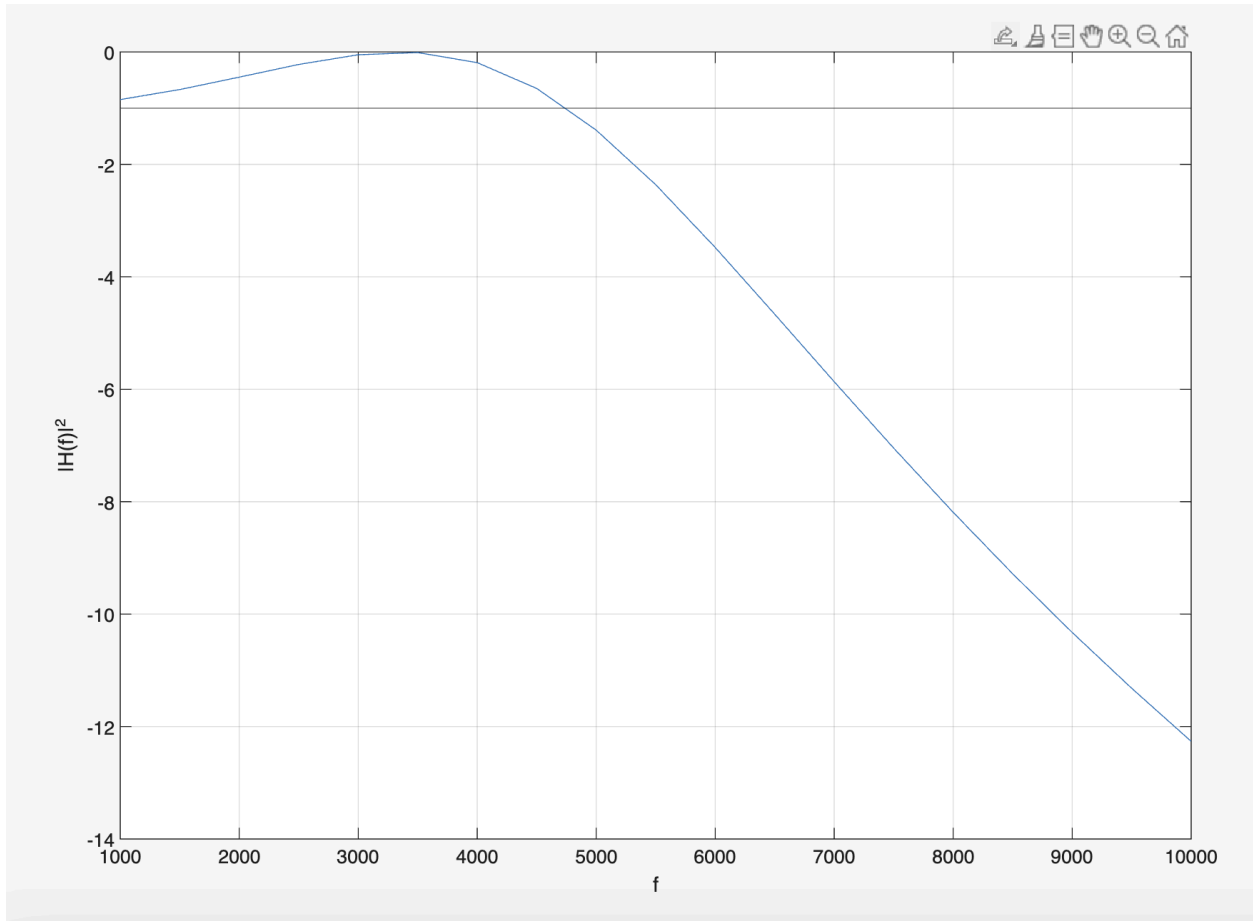


Figure 2: Matlab plot of $|H(f)|^2$ (dB) approximating f_o by setting it equal to 5000 Hz

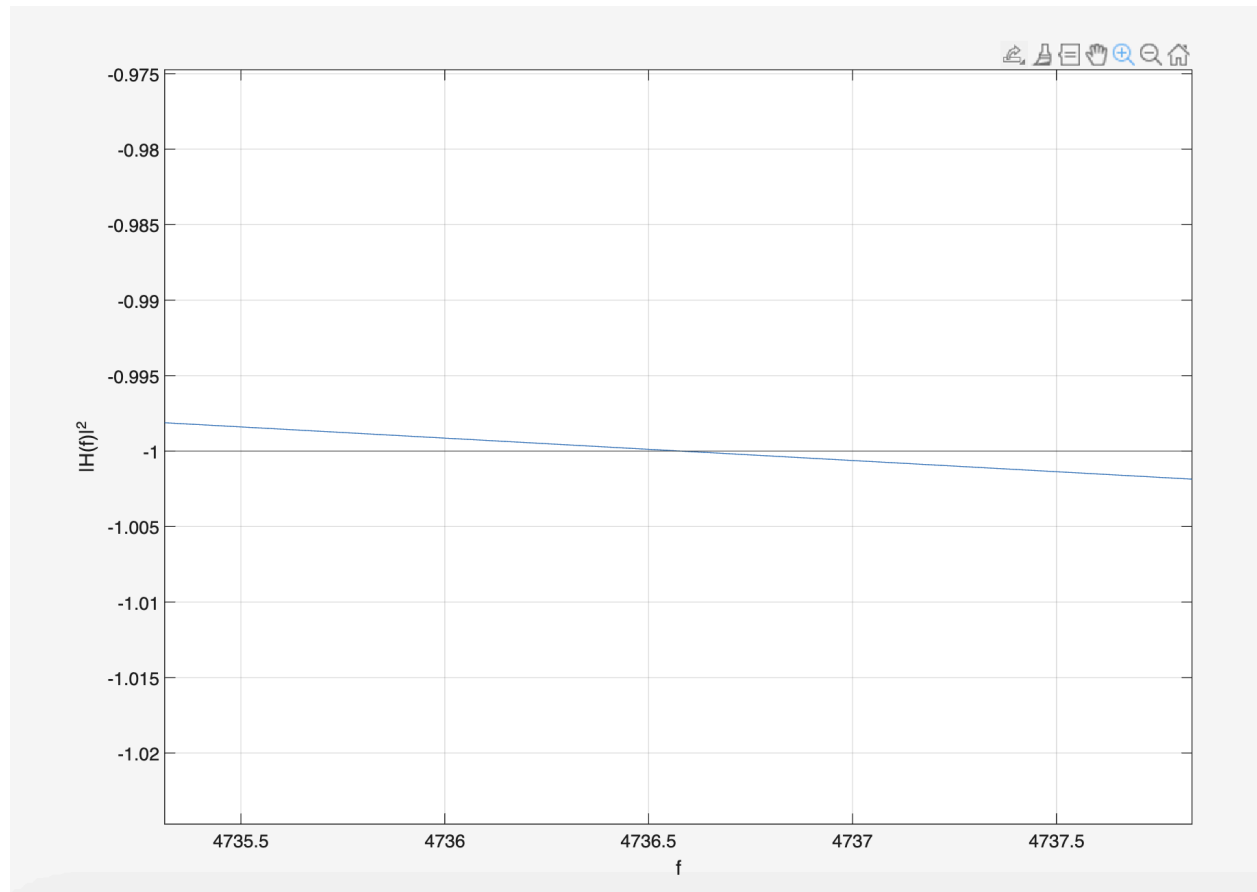


Figure 3: Figure 2 Enhanced showing frequency required to make transfer function -1dB with approximated f_o of 5000Hz

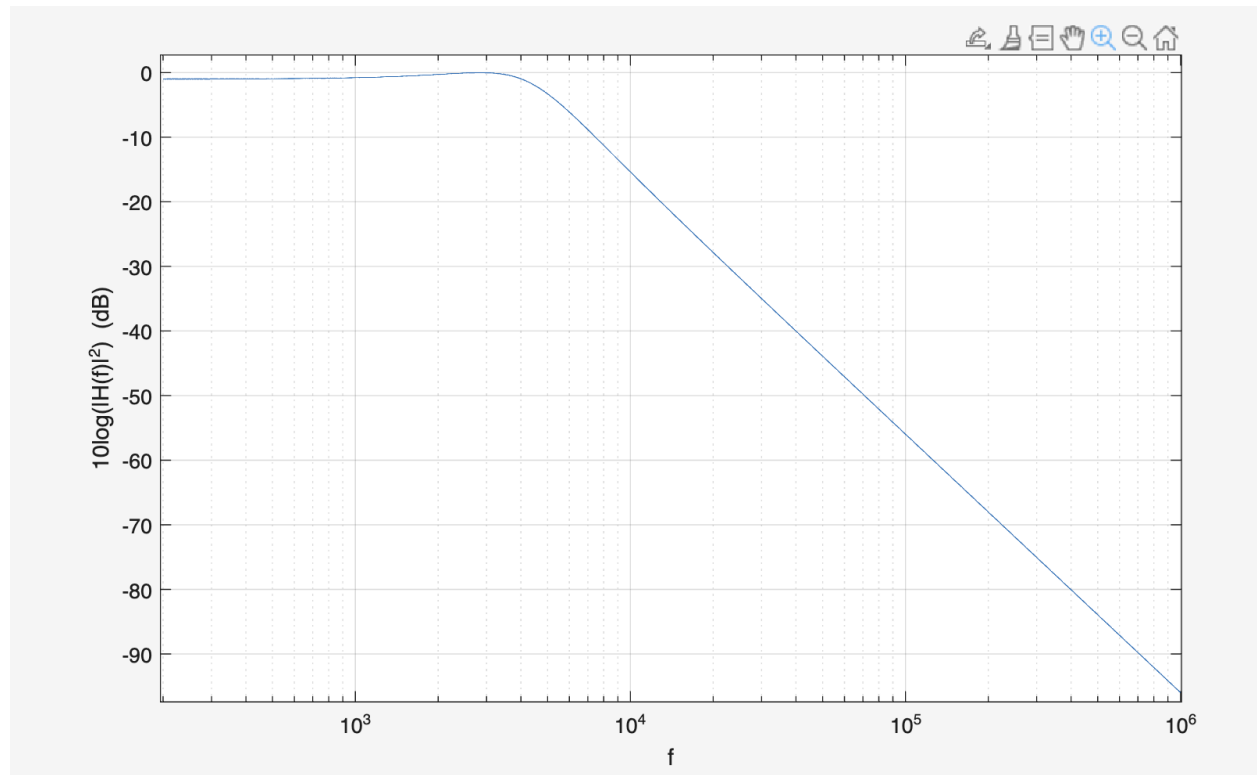


Figure 4: $|H(f)|^2$ in dB vs f for $200 < f < 1\text{MHz}$ on a semilogx plot

(d) Finally we need to adjust f_0 upwards a bit so $|H(f)|^2$ clears the passband edge and the stopband edge by the same factor. What is your final value of f_0 ? Save a copy of this graph to show that it meets the spec.

Hint: you can check the distance from the edge of the passband and the edge of the stopband in both frequency and gain.

Option 1: check the distance in gain	Option 2: check the distance in frequency
(-1dB) - Gain @4kHz = X1	(f @gain = -1 dB)/4 kHz = X1
Gain @32kHz - (-30dB) = X2	32 kHz/(f @gain = -30 dB) = X2
$ X1-X2 < 0.1dB$ is acceptable	$\frac{ X1-X2 }{X1} < 10\%$ is acceptable

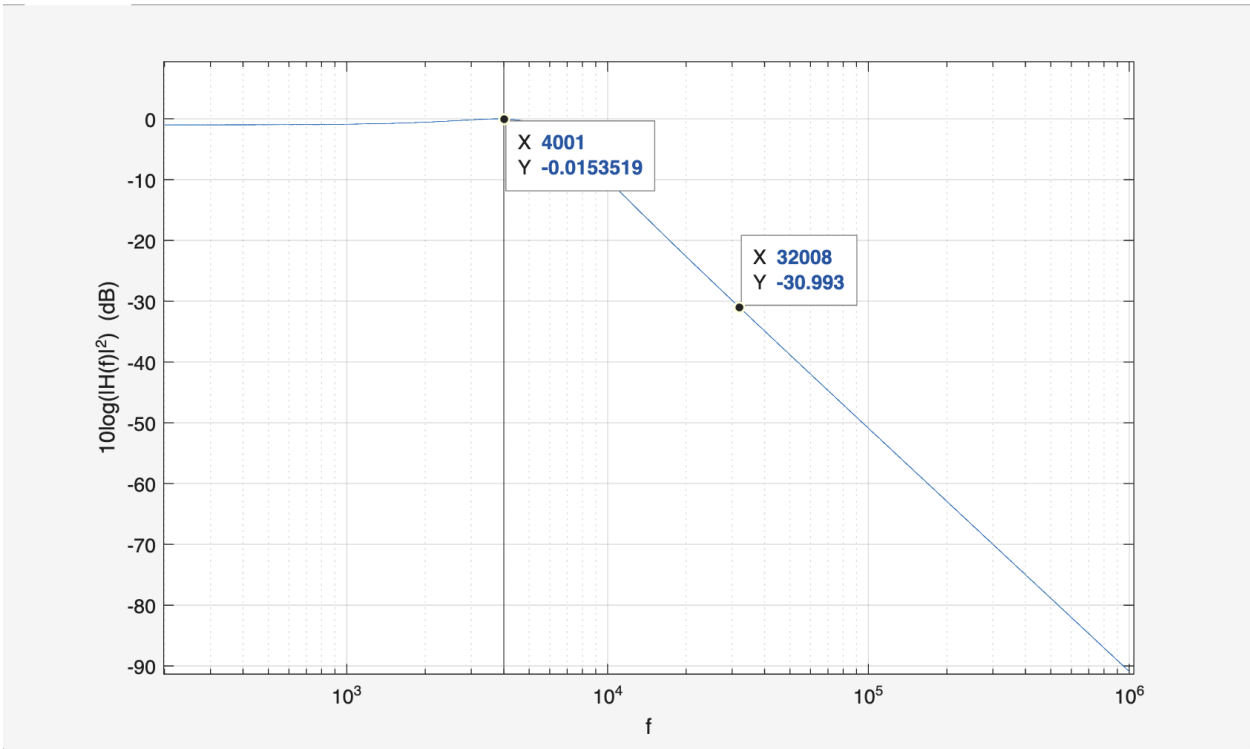


Figure 5: Adjusted f_0 (5655Hz) $|H(f)|^2$ clearing the passband edge and the stopband edge by the same factor Matlab Plot.

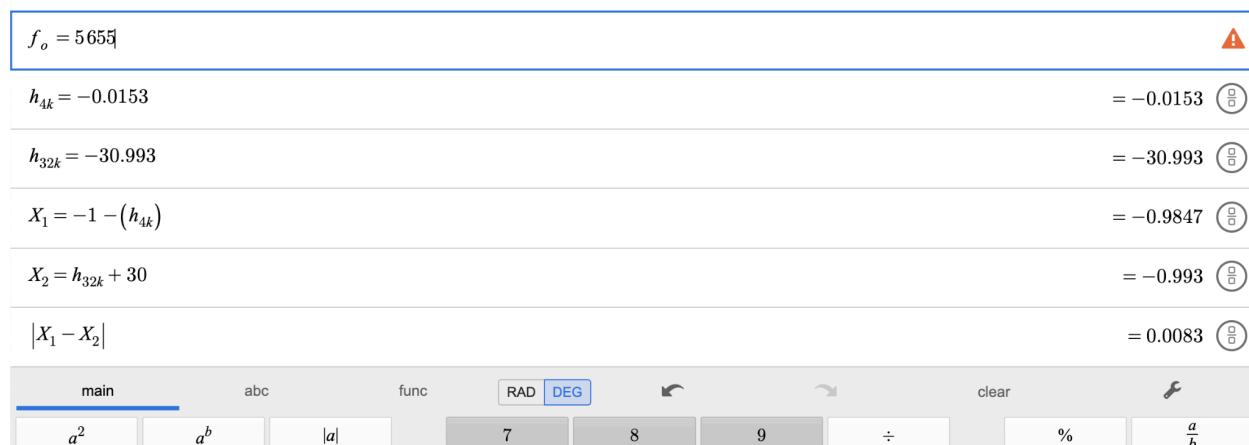


Figure 6: Adjusted f_o (5655Hz) $|H(f)|^2$ clearing passband edge calculation passing option 1

Part 2- Realization/Simulation

(a) Analyze the Sallen-Key circuit as shown on the following page. Put the transfer function

in the general form. For simplicity we usually set $R1 = R2$. Assume that the resistors are $R1 = R2 = 100K$. Calculate the component values needed to obtain the necessary values of f_0 and ζ . You found the necessary f_0 and ζ in parts 1(d) and 1(b), respectively. Finally we can correct the dc gain, setting it to -1dB, by converting R2 into a voltage divider with a dc gain of -1dB and a Thevenin resistance of 100K. Do this too.

Lab # 2

2)

$i_+ = i_- = 0$
 $V_+ = V_-$

KCL:

$$\frac{V_i - V_x}{Z_1} = \frac{V_x - V_o}{Z_3} + \frac{V_x - V_+}{Z_2}$$

$$\frac{V_i}{Z_1} - \frac{V_x}{Z_1} = \frac{V_x - V_o}{Z_3} + \frac{V_x - V_+}{Z_2}$$

$$\frac{V_i}{Z_1} = V_x \left(\frac{1}{Z_1} + \frac{1}{Z_3} + \frac{1}{Z_2} \right) - V_o \left(\frac{1}{Z_3} + \frac{1}{Z_2} \right)$$

KCL:

$$\frac{V_x - V_+}{Z_2} = \frac{V_+}{Z_4}$$

$$\frac{V_x}{Z_2} = \left(\frac{1}{Z_2} + \frac{1}{Z_4} \right) V_+$$

$$V_x = \left(1 + \frac{Z_2}{Z_4} \right) V_+$$

$$V_x = \left(1 + \frac{Z_2}{Z_4} \right) V_o$$

Substituting V_x into the first equation:

$$\frac{V_o}{V_i} = \frac{1}{1 + \frac{Z_1 + Z_2}{Z_4} + \frac{Z_1 Z_2}{Z_3 Z_4}}$$

Transfer function:

$$H(s) = \frac{1}{1 + \frac{R_2 + R_1}{sC_1} + \frac{(R_2)(R_1)}{\left(\frac{1}{sC_2}\right)\left(\frac{1}{sC_1}\right)}} = \frac{1}{1 + sC_1(R_1 + R_2) + s^2(R_1 R_2 C_1 C_2)}$$

$$H(j\omega) = \frac{1}{1 + j\omega C_1 (R_1 + R_2) + (j\omega)^2 (R_1 R_2 C_1 C_2)}$$

$$H(j\omega) = \frac{1}{1 + j 2 \frac{\omega}{\omega_0} + \frac{(j\omega)^2}{\omega_0^2}}$$

$$R_1 = R_2 = 100 \text{ k}$$

$$\omega_0 = 5655 \text{ Hz}$$

$$\xi = 0.523$$

$$\omega_0 = 2\pi(5655) \\ = 35531.41$$

$$2 \frac{\omega}{\omega_0} = C_1 (R_1 + R_2)$$

$$2 \frac{\omega}{\omega_0} = 2 R C_1$$

$$C_1 = \frac{\omega}{\omega_0} \cdot \frac{1}{R}$$

$$= \frac{0.523}{35531.41} \cdot \frac{1}{100 \text{ k}}$$

$$= 1.47 \cdot 10^{-10} \text{ F}$$

$$\text{pico F}$$

$$\frac{1}{\omega_0^2} = R_1 R_2 C_1 C_2$$

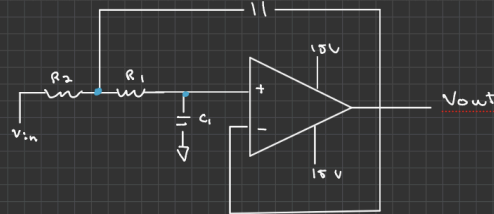
$$C_1 C_2 = \frac{1}{\omega_0^2} \cdot \frac{1}{R^2}$$

$$C_2 = \frac{1}{35531.41^2} \cdot \frac{1}{100^2} \cdot \frac{1}{1.47 \cdot 10^{-10}}$$

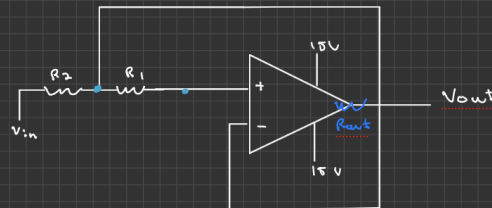
$$= 5.39 \cdot 10^{-10} \text{ F}$$

- (b) The most important “non-ideal” opamp effect for this circuit is the output resistance of the opamp itself. This is typically about 50Ω . At the highest frequencies you can assume that the capacitors are short circuits and the opamp gain is zero. Show that under these conditions $|H| \rightarrow \frac{R_{OUT}}{R}$. (Assume $R_1 = R_2 = R$ in the schematic shown below.)

b)



@ higher frequencies : capacitors short



gain = 0

circuit reduces to voltage divider

$$H = \frac{V_o}{V_i} = \frac{R \parallel R_{out}}{R + (R \parallel R_{out})} = \frac{\frac{R \cdot R_{out}}{R + R_{out}}}{R + \frac{R \cdot R_{out}}{R + R_{out}}} = \frac{R_{out}}{R + 2R_{out}}$$

since $R_{out} \ll R$

$$H \approx \frac{R_{out}}{R}$$

$$|H| = \frac{50}{100k} = 5e^{-4} \rightarrow 20 \log(5e^{-4}) = -66 \text{ dB}$$

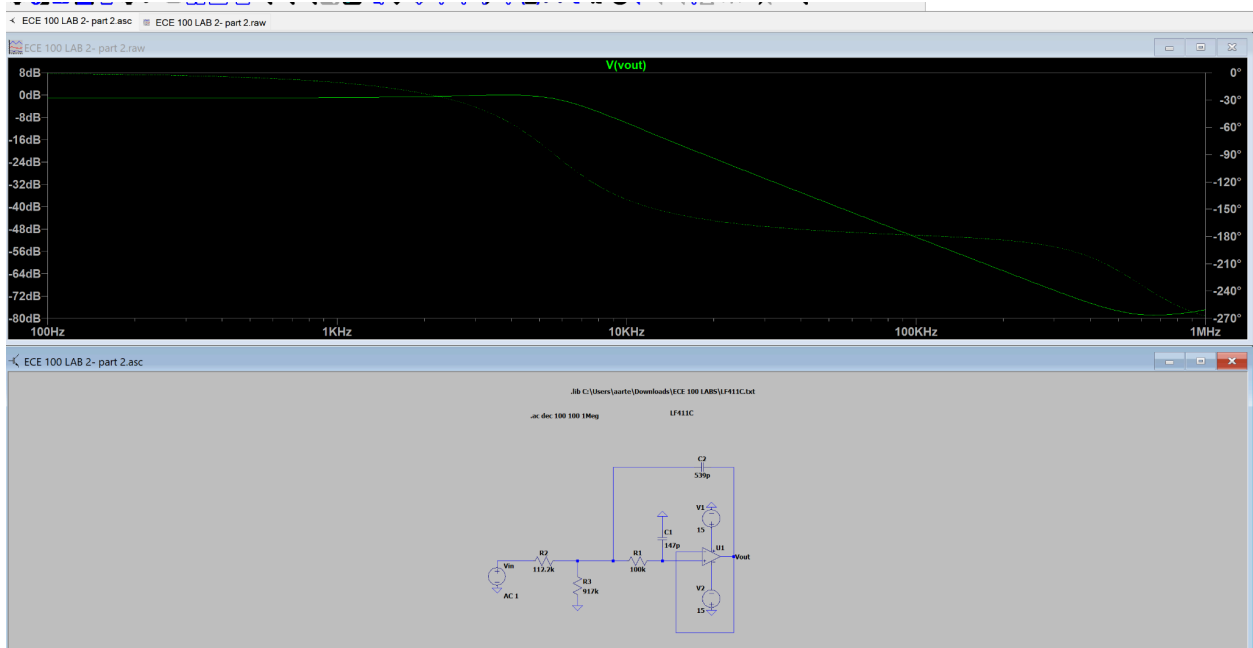


Figure 7: Simulation of circuit with LF411 op-amp

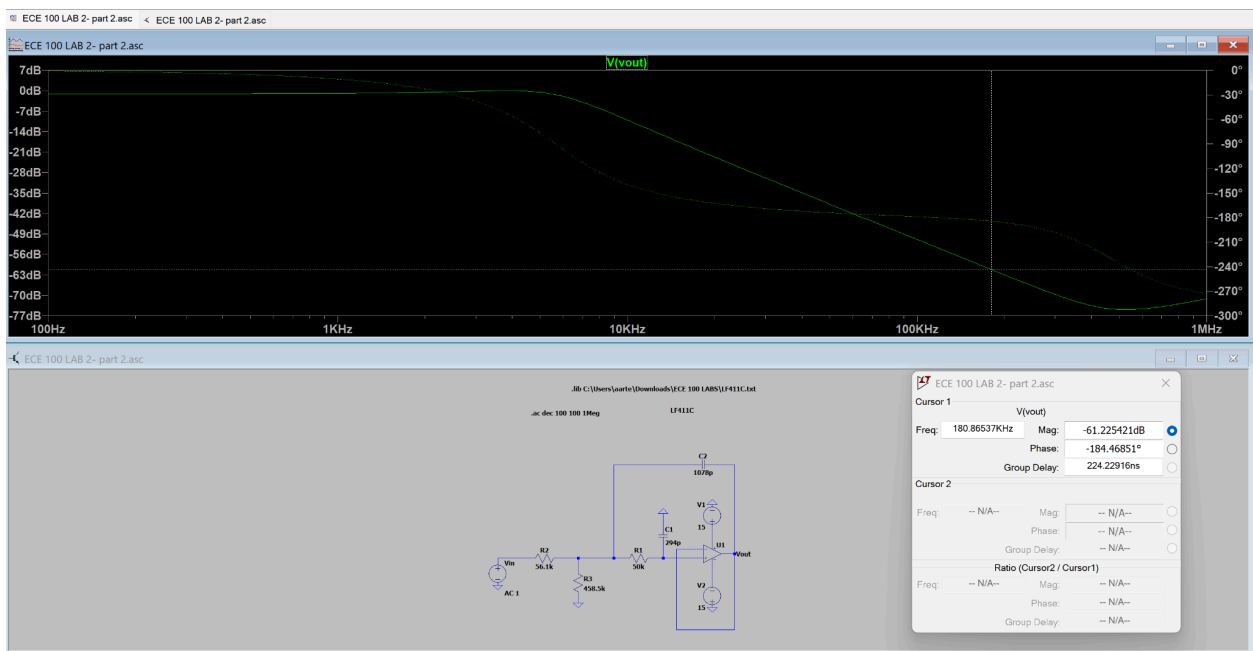


Figure 8: Simulation of circuit with LF411 op-amp with resistors halved and capacitor values double

The simulated transfer function still met the project specifications. The passband and transition-band behavior remained essentially unchanged, and the high-frequency attenuation requirement was still satisfied. From the LTspice simulation, the gain at 180.87 kHz was about

−61.23 dB, which is below the required −55 dB, and the response stayed below −55 dB from 180 kHz to 1 MHz.

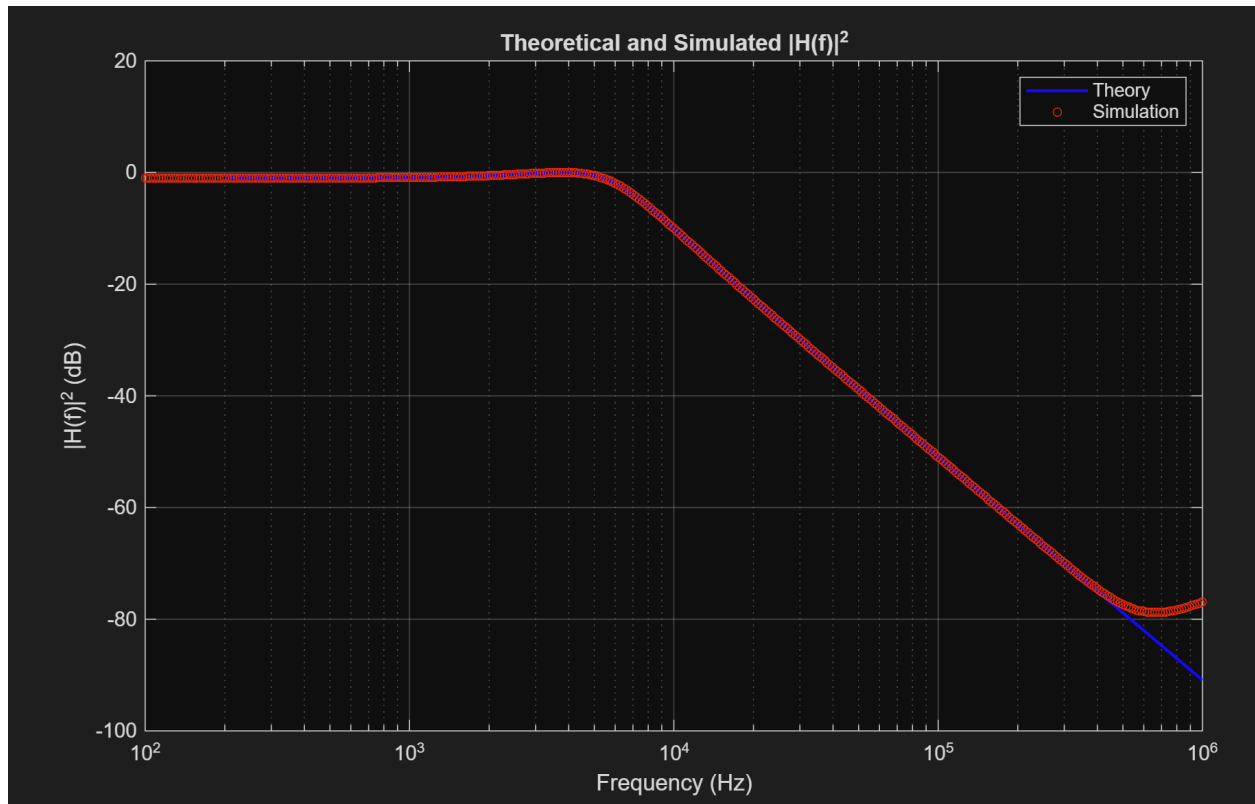


Figure 8: Theoretical and simulated data sets of the squared magnitude response

Part 3 - Experiment

3. **Experiment:** Build and test the active circuit. We will use the LF411 op-amp with the pins shown below. In our lab, we have lots of 1% resistors in stock so you should be able to get pretty close to the three resistors you need. You can always use a resistor larger than the minimum value determined in your simulation to make the choice easier. However we don't have nearly as wide a range of capacitors, you will probably have to put two capacitors in parallel to approximate C1 and the same for C2.

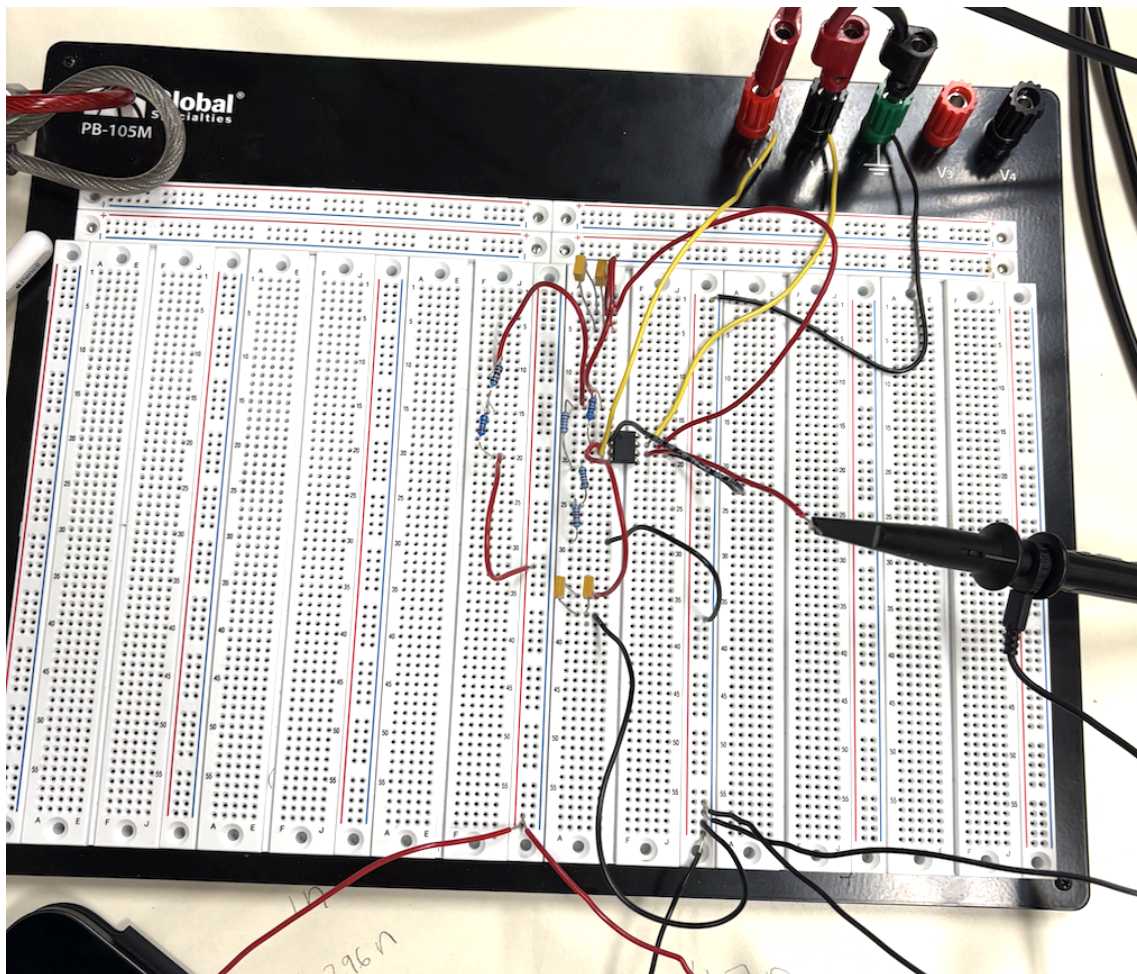


Figure 9: Sallen-Key Circuit

- (a) Put probes on both the input and the output and trigger a sine wave form using signal generator at the input probe. This will allow you to measure low-level signals at the output without losing the trigger. Measure the voltage gain at the spec frequencies: DC(use 5hz or 10Hz instead of 0Hz), fPEAK, 4 Khz, 32 Khz, 180 Khz, and 1 Mhz. Create a table to report your measured input and output peak-to-peak voltage amplitudes and the calculated voltage gain in dB.

$$V_{peak} = V_o \sqrt{1 + 2f^2}$$

$$f_{peak} = 5655 \text{ Hz} \sqrt{1 + 2(0.523)^2}$$

$$= 7033 \text{ Hz}$$

10Vpk-pk	DC (0-5)Hz	F_peak 7033Hz	4kHz	32kHz	180kHz	1MHz
V_i	10.2V	10.7V	10.7V	10.7V	10.3V	19.77
V_o	9.2V	5V	9.6V	237mV	30mV	80mV
Gain (dB)	-0.9809879 472	-5.3	-0.8522408 684	-33.0927	-50.153744 93	-47.871504 07

Gain tapers off at 32kHz and is no longer spec (higher by ~5dB) this is likely because of the compounding percentages of accuracy from the resistors and capacitors because we used multiple and also because we could not get exact resistor and cap values

- (b) Explore the remainder of the frequency range up to the generator limit and make sure there are no unexpected “features.” Since this is an active circuit you have to make sure that the signal does not drive the op-amp into a nonlinear operating condition. So long as the output voltage still looks like a sine wave you are probably OK. A good way to confirm that the

system is being tested in a linear range is to change the input voltage by a factor of two and re-measure $|H(f)|^2$, if it is the same you are OK. Be sure to take a scope image before and after changing the input voltage. If they differ then reduce the input voltage by another factor of two and try again. Put your measurements in an ascii file so you can read them into MATLAB and plot them on top of the simulation.

10Vpk-pk	5MHz	10MHz	15MHz	20MHz
V _i (10V)	7.7V	5.02V	3.76V	3.02V
V _o	136mV	150mV	140mV	102mV
Gain (dB)	-35.059036 34	-30.5095345 2	-28.558064 64	-29.370421 66

10Vpk-pk	5MHz	10MHz	15MHz	20MHz
V _i (20)	15.13V	9.93V	7.45V	5.96V
V _o	278mV	300mV	291mV	217.2mV
Gain (dB)	-34.698643 03	-30.396559 88	-28.106774 6	-28.767728 78

Additional measurements were taken in the MHz range to further evaluate high-frequency attenuation and confirm op-amp non-ideal behavior.

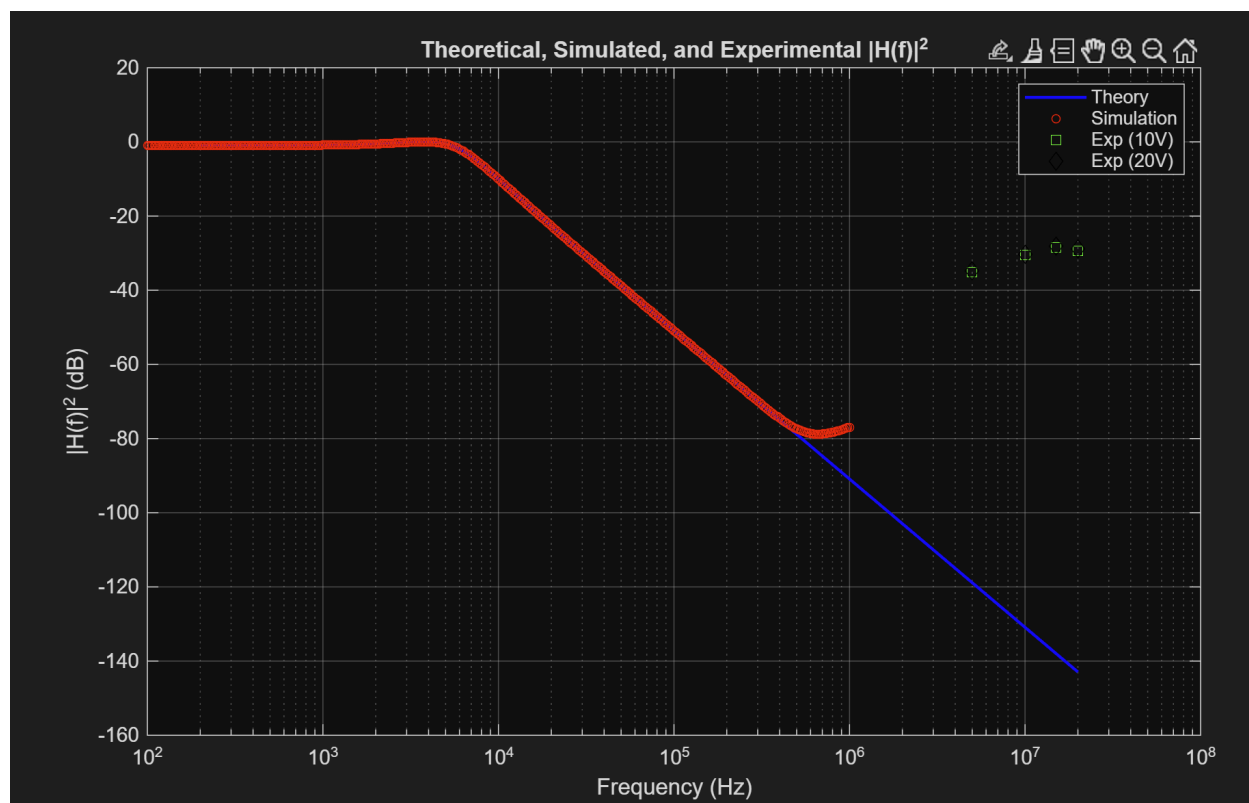


Figure 10: Matlab Theoretical simulation with Experimental Overlay (part b data)

```

>>| % Given parameters
f0 = 5655;
zeta = 0.523;

% =====
% THEORETICAL CURVE
% =====
f = logspace(log10(200), log10(2e7), 3000); % extended to 20 MHz

H2_theory = 0.7943 ./ ((1 - (f./f0).^2).^2 + (2*zeta*(f./f0)).^2);
H2_theory_dB = 10*log10(H2_theory);

% =====
% SIMULATION DATA (LTSpice)
% =====
Hsim = Vre + 1j*Vim; % assumes already imported
H2_sim_dB = 10*log10(abs(Hsim).^2); % magnitude squared in dB

% =====
% EXPERIMENTAL DATA
% =====
f_exp = [5e6, 10e6, 15e6, 20e6];

gain_exp_10 = [-35.05903634, -30.50953452, -28.55806464, -29.37042166];
gain_exp_20 = [-34.69864303, -30.39655988, -28.1067746, -28.76772878];

% =====
% PLOTTING
% =====
figure

% Theory
semilogx(f, H2_theory_dB, 'b-', 'LineWidth', 1.5)
hold on

```

```

% Simulation
semilogx(Freq, H2_sim_dB, 'ro', 'MarkerSize', 4)

% Experimental (10V)
semilogx(f_exp, gain_exp_10, 'gs', 'MarkerSize', 6)

% Experimental (20V)
semilogx(f_exp, gain_exp_20, 'kd', 'MarkerSize', 6)

grid on
xlabel('Frequency (Hz)')
ylabel('|H(f)|^2 (dB)')
title('Theoretical, Simulated, and Experimental |H(f)|^2')

legend('Theory', 'Simulation', 'Exp (10V)', 'Exp (20V)')
>>

```

Conclusion

The calculated, simulated, and experimental results represent how closely the designed active low pass filter satisfied the required frequency response specifications for DSL and voice signal separation. The analytical calculations established the needed transfer function shape, damping factor, and natural frequency, while the MATLAB plots showed how allowing a small amount of passband ripple improved the transition band performance compared with the Butterworth case and gave more clearance at the stopband edge. The Sallen Key realization translated those target parameters into resistor and capacitor values, and the LF411 simulation showed that non-ideal op amp behavior becomes more important at high frequencies and can affect whether the design still meets the specification. The experimental measurements at dc, f_{peak}, 4 kHz, 32kHz, 180kHz, and 1MHz provided a direct physical interpretation of the

calculated gains, showing whether the real circuit preserved the expected passband level, transition behavior, and high frequency attenuation. In general, there was mostly reasonable agreement among the simulated, analytical design, and measured results, with the biggest takeaway being that successful filter design requires deriving the ideal transfer function and accounting for implementation and measurement effect so the final circuit still meets required practice specifications.

Matlab Code

```
%Timothy Carroll A19112628
clear; clc; close all;

z = 0.1:0.01:0.7;

h = 1 ./ (4*z.^2 .* (1 - z.^2));

hdb = 10*log10(h);

plot(z, hdb)
grid on
xlabel('\zeta')
ylabel('|H(f_{peak})|^2 / |H(0)|^2 (dB)')
hold on
yline(1);|
```

Matlab Code Block for Figure 1

```

1 %Timothy Carroll A19112628
2 clear; clc; close all;
3
4 f = 1000:500:10000;
5
6 h = 0.7943./((1-(f/(5000)).^2).^2+(2*0.523*f/(5000)).^2);
7
8
9 hdb = 10*log10(h);
10
11
12 plot(f, hdb)
13 grid on
14 xlabel('\zeta')
15 ylabel('|H(f_{peak})|^2 / |H(0)|^2 (dB)')
16 hold on
17 yline(-1);
18

```

Matlab Code Block for Figure 2

F appears to be 4736.5 but after doing conversion $f_o = 4222.53\text{Hz}$

```

1 %Find the required f0 and plot |H(f)|^2 in
2 %dB vs f for 200 < f < 1MHz on a semilogx plot.
3
4 %Timothy Carroll A19112628
5 clear; clc; close all;
6
7 f = 200:1e6;
8
9 h = 0.7943./((1-(f/(4222.53)).^2).^2+(2*0.523*f/(4222.53)).^2);
10
11
12 hdb = 10*log10(h);
13
14
15 semilogx(f, hdb)
16 grid on
17 xlabel('f')
18 ylabel('10log(|H(f)|^2) (dB)')
19 hold on
20

```

Matlab Code Block for Figure 4

```

1 %Find the required f0 and plot |H(f)|^2 in
2 %dB vs f for 200 < f < 1MHz on a semilogx plot.
3
4 %Timothy Carroll A19112628
5 clear; clc; close all;
6
7 f = 200:1e6;
8
9 h = 0.7943./((1-(f/(5655)).^2).^2+(2*0.523*f/(5655)).^2);
10
11
12 hdb = 10*log10(h);
13
14
15 semilogx(f, hdb)
16 grid on
17 xlabel('f')
18 ylabel('10log(|H(f)|^2) (dB)')
19 hold on
20 xline(4000);
21

```

Matlab Code Block for Figure 5

```

>> f0 = 5655;
zeta = 0.523;

% theoretical curve
f = logspace(log10(200), log10(1e6), 2000);
H2_theory = 0.7943 ./ ((1 - (f./f0).^2).^2 + (2*zeta*(f./f0)).^2);
H2_theory_dB = 10*log10(H2_theory);

% simulated data imported from LTspice
Hsim = Vre + 1j*Vim;
H2_sim_dB = 10*log10(abs(Hsim).^2);

figure
semilogx(f, H2_theory_dB, 'b-', 'LineWidth', 1.5)
hold on
semilogx(Freq, H2_sim_dB, 'ro', 'MarkerSize', 4)

grid on
xlabel('Frequency (Hz)')
ylabel('|H(f)|^2 (dB)')
title('Theoretical and Simulated |H(f)|^2')
legend('Theory','Simulation')
>>

```

Figure : Theoretical and simulated data for the squared magnitude of the frequency response

```

>>| % Given parameters
f0 = 5655;
zeta = 0.523;

% =====
% THEORETICAL CURVE
% =====
f = logspace(log10(200), log10(2e7), 3000); % extended to 20 MHz

H2_theory = 0.7943 ./ ((1 - (f./f0).^2).^2 + (2*zeta*(f./f0)).^2);
H2_theory_dB = 10*log10(H2_theory);

% =====
% SIMULATION DATA (LTSpice)
% =====
Hsim = Vre + 1j*Vim; % assumes already imported
H2_sim_dB = 10*log10(abs(Hsim).^2); % magnitude squared in dB

% =====
% EXPERIMENTAL DATA
% =====
f_exp = [5e6, 10e6, 15e6, 20e6];

gain_exp_10 = [-35.05903634, -30.50953452, -28.55806464, -29.37042166];
gain_exp_20 = [-34.69864303, -30.39655988, -28.1067746, -28.76772878];

% =====
% PLOTTING
% =====
figure

% Theory
semilogx(f, H2_theory_dB, 'b-', 'LineWidth', 1.5)
hold on

```

Figure 10 Matlab Code

```

% Simulation
semilogx(Freq, H2_sim_dB, 'ro', 'MarkerSize', 4)

% Experimental (10V)
semilogx(f_exp, gain_exp_10, 'gs', 'MarkerSize', 6)

% Experimental (20V)
semilogx(f_exp, gain_exp_20, 'kd', 'MarkerSize', 6)

grid on
xlabel('Frequency (Hz)')
ylabel('|H(f)|^2 (dB)')
title('Theoretical, Simulated, and Experimental |H(f)|^2')

legend('Theory', 'Simulation', 'Exp (10V)', 'Exp (20V)')

```

Figure 10 Matlab code Cont.